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<u>Written Analysis</u>

Python Model

Data Transformation

Extracted Year, Month, and Hour categorical variables from the Time series feature to improve usability in ML models. These had chi square values of 141.64, 167.18, and 34.54, respectively. Additionally, having noticed that all anomalies took place in the first half of the day (between 0 and 13 hours), I created a boolean variable called hour_0_thru_13 that reports whether the hour is in that range or not. Features like Z-scores ultimately provided no additional value as the correlation coefficients were the same as the column in which the z-score was calculated off. Initially, I thought calculating the distance between the sum_amount and avg_amount_per_user would be useful. However, it yielded a correlation coefficient of -0.0051, so I did not include it in the final model.

Training

First, I encoded the categorical variables. Then I dropped the columns I had deemed to be not useful. I noticed the models performed better when the test portion of the train-test split, and so I decided to go with 60% train-40% test. I had planned to stratify this due to the small portion of anomalies in the context of the entire dataset, however although it was working at first at the end it was giving me an error so I dropped it.

Evaluation

Ultimately KNN performed the best, with an F1 of 0.914 and a balanced accuracy of 0.944. Below are the Precision-Recall and ROC curves showing model performance.

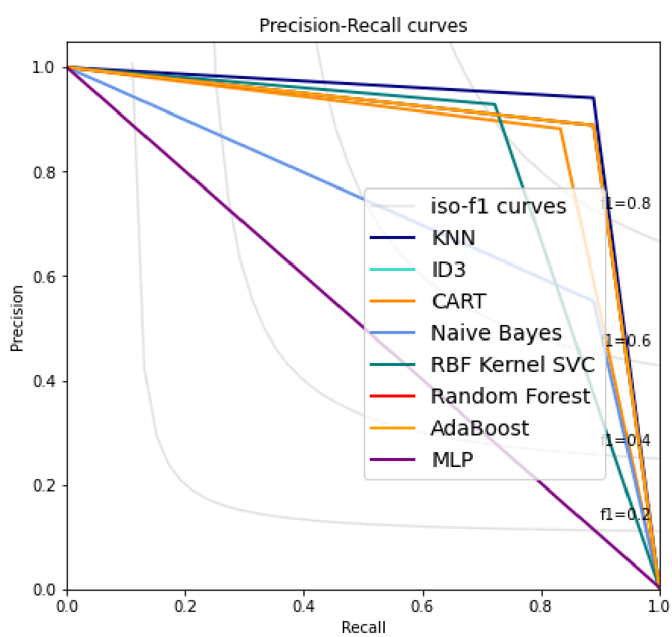

Figure 1. Precision-Recall curve

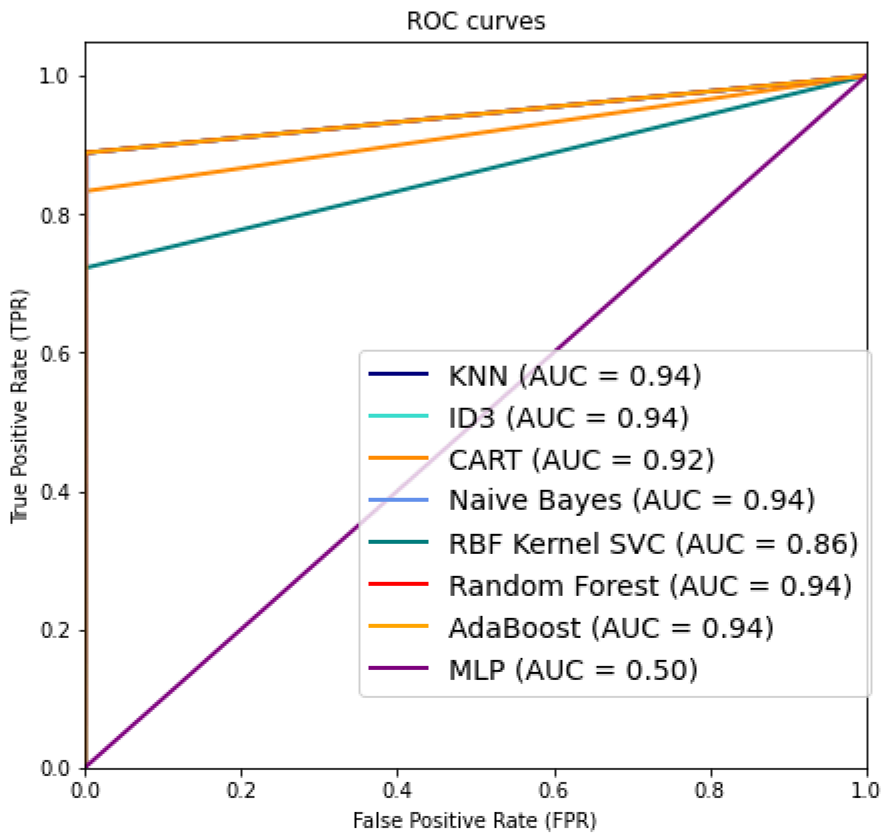

Figure 2. ROC curve